

# H524 Introduction to Biostatistics

## Week 10: More on Simple Linear Regression & Project Presentations

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# Outline

- 1 Review from Week 9
- 2 Regression Assumptions
- 3 Diagnostic Plots
- 4 Outliers and Influential Points
- 5 Transformations
- 6 Hypothesis Testing and Confidence Intervals
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# Quick Review: Simple Linear Regression

## Last week we covered:

- Simple linear regression model:  $Y = \beta_0 + \beta_1 X + \epsilon$
- Least squares estimation: minimizing residuals
- Coefficient interpretation
- Correlation vs. regression
- R-squared and model fit

**This Week:** Model diagnostics, assumptions, transformations, and presenting results

**Reading:** Pagano & Gauvreau Chapter 18 (continued)

# The Fitted Model Recap

## Simple Linear Regression:

$$\hat{y} = b_0 + b_1x$$

where:

- $b_1 = r \cdot \frac{s_y}{s_x}$  (slope)
- $b_0 = \bar{y} - b_1\bar{x}$  (intercept)
- $r$  = correlation coefficient

## Key Concepts:

- **Residual:**  $e_i = y_i - \hat{y}_i$  (observed - predicted)
- **Sum of Squared Errors:**  $SSE = \sum e_i^2$
- **R-squared:**  $R^2 = r^2$  (proportion of variance explained)

# The Four Key Assumptions

**For valid inference from linear regression:**

- ➊ **Linearity:** The relationship between  $X$  and  $Y$  is linear
  - Check: Scatterplot, residual plot
- ➋ **Independence:** Observations are independent
  - Check: Study design, temporal patterns
- ➌ **Homoscedasticity:** Constant variance of residuals
  - Check: Residual vs fitted plot
- ➍ **Normality:** Residuals are approximately normal
  - Check: Q-Q plot, histogram of residuals

**Acronym:** LINE (Linearity, Independence, Normality, Equal variance)

# Why Assumptions Matter

## Consequences of violated assumptions:

### Linearity violation:

- Biased estimates
- Poor predictions
- Misleading relationships

### Independence violation:

- Underestimated standard errors
- Invalid p-values and confidence intervals
- False sense of precision

### Homoscedasticity violation:

- Inefficient estimates
- Invalid hypothesis tests
- Incorrect prediction intervals

### Normality violation:

# Residual vs Fitted Plot

**Purpose:** Check linearity and homoscedasticity

**What to look for:**

- **Good:** Random scatter around horizontal line at zero
- **Bad:** Patterns, curves, funneling

**Common patterns:**

- **Curved pattern:** Non-linear relationship (try transformation)
- **Funnel shape:** Non-constant variance (heteroscedasticity)
- **Clusters/groups:** Missing categorical variable
- **Outliers:** Points far from the cloud

**R code:**

```
plot(fitted(model), residuals(model))  
abline(h=0, lty=2)
```

# Q-Q Plot (Quantile-Quantile)

**Purpose:** Check normality of residuals

**What to look for:**

- **Good:** Points fall approximately on a straight line
- **Bad:** Systematic departures from the line

**Common patterns:**

- **S-shaped curve:** Heavy tails (outliers)
- **Curve below then above:** Right skewness
- **Curve above then below:** Left skewness
- **Points at ends deviate:** Extreme values

**R code:**

```
qqnorm(residuals(model))  
qqline(residuals(model))
```

**Note:** Minor departures are okay, especially with small samples



# Scale-Location Plot

**Purpose:** Check homoscedasticity (constant variance)

**What it shows:** Square root of standardized residuals vs fitted values

**What to look for:**

- **Good:** Random scatter with horizontal trend line
- **Bad:** Upward or downward trend

**Interpretation:**

- Horizontal line  $\Rightarrow$  constant variance
- Upward slope  $\Rightarrow$  variance increases with fitted values
- Downward slope  $\Rightarrow$  variance decreases with fitted values

**R code:**

```
plot(model, which=3)
```

# Residuals vs Leverage Plot

**Purpose:** Identify influential observations

## Key Concepts:

- **Leverage:** How far  $x_i$  is from  $\bar{x}$  (potential influence)
- **Cook's Distance:** Actual influence on fitted values
- **Influential point:** High leverage AND large residual

## What to look for:

- Points outside Cook's distance contours (dashed lines)
- Cook's distance  $> 0.5$  or  $> 1$  suggests influential observation

## R code:

```
plot(model, which=5)
```

**Action:** Investigate influential points, but don't automatically remove them

# Creating All Diagnostic Plots in R

## Method 1: All four plots at once

```
par(mfrow=c(2,2))  
plot(model)  
par(mfrow=c(1,1))
```

## Method 2: Individual plots

```
# 1. Residuals vs Fitted
```

```
plot(model, which=1)
```

```
# 2. Q-Q plot
```

```
plot(model, which=2)
```

```
# 3. Scale-Location
```

```
plot(model, which=3)
```

```
# 4. Residuals vs Leverage
```

```
plot(model, which=5)
```

# Types of Unusual Observations

## Three types to distinguish:

### 1. Outlier:

- Large residual (far from regression line)
- May or may not be influential
- Check for data entry errors

### 2. High leverage point:

- Extreme  $x$  value (far from  $\bar{x}$ )
- Potential to be influential
- May not have large residual

### 3. Influential point:

- High leverage AND large residual
- Removing it substantially changes regression line
- Check Cook's distance

## DO NOT automatically remove outliers!

### Steps to take:

#### 1 Check for errors:

- Data entry mistakes
- Measurement errors
- Wrong units

#### 2 Investigate context:

- Is it a legitimate observation?
- Does it represent a special subgroup?
- Was there a protocol violation?

#### 3 Perform sensitivity analysis:

- Fit model with and without the point
- Report both analyses
- Discuss impact on conclusions

#### 4 Consider robust methods:

- Robust regression (less sensitive to outliers)
- Transformations

## Example: Identifying Influential Points in R

```
# Fit model
model <- lm(y ~ x, data=mydata)

# Calculate Cook's distance
cooks <- cooks.distance(model)

# Identify influential points (Cook's D > 0.5)
influential <- which(cooks > 0.5)
print(influential)

# Show influential observations
mydata[influential, ]

# Compare models with/without influential points
model_full <- lm(y ~ x, data=mydata)
model_reduced <- lm(y ~ x, data=mydata[-influential,])

summary(model_full)
summary(model_reduced)
```

# When to Transform Data

## Transformations can help when:

- Relationship is non-linear
- Variance is non-constant
- Residuals are non-normal
- Interpretation requires it (e.g., multiplicative effects)

## Common transformations:

- **Log:**  $\log(Y)$  or  $\log(X)$  – for right skewness, exponential growth
- **Square root:**  $\sqrt{Y}$  – for count data, mild right skewness
- **Reciprocal:**  $1/Y$  or  $1/X$  – for rates, strong right skewness
- **Square:**  $Y^2$  or  $X^2$  – for quadratic relationships

**Box-Cox transformation:** Data-driven choice of power transformation

## Most common transformation in biostatistics

### Model forms:

- ➊ **Log-linear:**  $\log(Y) = \beta_0 + \beta_1 X$ 
  - Interpretation: 1-unit increase in  $X \Rightarrow (e^{\beta_1} - 1) \times 100\%$  change in  $Y$
- ➋ **Linear-log:**  $Y = \beta_0 + \beta_1 \log(X)$ 
  - Interpretation: 1% increase in  $X \Rightarrow \beta_1/100$  unit change in  $Y$
- ➌ **Log-log:**  $\log(Y) = \beta_0 + \beta_1 \log(X)$ 
  - Interpretation:  $\beta_1$  is the elasticity (1% change in  $X \Rightarrow \beta_1\%$  change in  $Y$ )

### When to use:

- Right-skewed data
- Multiplicative relationships
- Relative changes more meaningful than absolute



# Example: Log Transformation in R

## Scenario: BMI vs medical cost (right-skewed)

# Original model (may violate assumptions)

```
model_original <- lm(cost ~ bmi, data=health)
```

# Log-transformed response

```
model_log <- lm(log(cost) ~ bmi, data=health)
```

# Compare diagnostics

```
par(mfrow=c(2,2))
```

```
plot(model_original)
```

```
plot(model_log)
```

```
par(mfrow=c(1,1))
```

# Interpretation

```
summary(model_log)
```

# If  $\beta_1 = 0.05$ , then 1-unit increase in BMI

# increases cost by  $(\exp(0.05)-1)*100 = 5.1\%$

# Polynomial Regression

## For non-linear relationships with turning points

### Quadratic model:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$$

### When to use:

- U-shaped or inverted U-shaped relationship
- Single peak or valley
- Example: age vs bone density (increases then decreases)

### Interpretation:

- $\beta_1$ : slope at  $X = 0$
- $\beta_2$ : rate of change in slope
- If  $\beta_2 > 0$ : U-shaped (convex)
- If  $\beta_2 < 0$ : inverted U-shaped (concave)

**Warning:** Don't extrapolate beyond data range!

# Polynomial Regression in R

```
# Create quadratic term
mydata$age_squared <- mydata$age^2

# Fit quadratic model
model_quad <- lm(bone_density ~ age + age_squared,
                 data=mydata)

# Alternative: use poly() function
model_quad2 <- lm(bone_density ~ poly(age, 2, raw=TRUE),
                 data=mydata)

summary(model_quad)

# Find peak (maximum bone density)
# Peak at: -beta1 / (2 * beta2)
beta1 <- coef(model_quad)[2]
beta2 <- coef(model_quad)[3]
peak_age <- -beta1 / (2 * beta2)
print(paste("Peak at age:", round(peak_age, 1)))
```

# Testing the Slope

**Null hypothesis:**  $H_0 : \beta_1 = 0$  (no linear relationship)

**Alternative:**  $H_A : \beta_1 \neq 0$  (linear relationship exists)

**Test statistic:**

$$t = \frac{b_1 - 0}{SE(b_1)} \sim t_{n-2}$$

where  $SE(b_1) = \frac{s_e}{\sqrt{\sum(x_i - \bar{x})^2}}$  and  $s_e = \sqrt{\frac{SSE}{n-2}}$

**Decision:**

- Reject  $H_0$  if  $|t| > t_{\alpha/2, n-2}$  or  $p < \alpha$
- Fail to reject if  $|t| \leq t_{\alpha/2, n-2}$  or  $p \geq \alpha$

**Note:** This tests whether  $X$  is useful for predicting  $Y$

# Confidence Interval for Slope

## 95% CI for $\beta_1$ :

$$b_1 \pm t_{0.975, n-2} \times SE(b_1)$$

## Interpretation:

- We are 95% confident the true slope is in this interval
- If interval excludes 0, reject  $H_0 : \beta_1 = 0$
- Width reflects precision of estimate

## Example:

- $b_1 = 2.5$  (estimated slope)
- $SE(b_1) = 0.8$
- $n = 30$ , so  $t_{0.975, 28} = 2.048$
- 95% CI:  $2.5 \pm 2.048(0.8) = (0.86, 4.14)$
- Interpretation: For each 1-unit increase in  $X$ ,  $Y$  increases between 0.86 and 4.14 units

# Confidence vs Prediction Intervals

## Two types of intervals for predictions:

### 1. Confidence Interval for Mean Response:

- Estimates the **average**  $Y$  for a given  $X = x_0$
- Answers: What is the mean outcome for this  $X$ ?
- Narrower interval
- Example: Mean blood pressure for all 50-year-olds

### 2. Prediction Interval for Individual Response:

- Predicts a **single**  $Y$  for a given  $X = x_0$
- Answers: What will this individual's outcome be?
- Wider interval (includes individual variability)
- Example: Blood pressure for one specific 50-year-old

**Key difference:** Prediction interval accounts for both:

- 1 Uncertainty in estimating the mean (same as CI)
- 2 Variability of individuals around the mean (extra)

# Computing Intervals in R

```
# Fit model
model <- lm(sbp ~ age, data=health)

# New data point
new_data <- data.frame(age = 50)

# Confidence interval for mean SBP at age 50
predict(model, newdata=new_data,
        interval="confidence", level=0.95)
# Example output: fit=130, lwr=128, upr=132

# Prediction interval for individual at age 50
predict(model, newdata=new_data,
        interval="prediction", level=0.95)
# Example output: fit=130, lwr=110, upr=150

# Note: Prediction interval is wider!
```

# Elements of a Good Statistical Presentation

**For your final projects, include:**

**1 Clear research question**

- What are you trying to learn?

**2 Descriptive statistics**

- Sample size, means, SDs
- Summary tables

**3 Visualizations**

- Scatterplots, histograms, boxplots
- Clear labels and legends

**4 Statistical methods**

- Which tests/models did you use?
- Why did you choose them?

**5 Results**

- Effect sizes, confidence intervals, p-values
- Interpretation in context

**6 Limitations and assumptions**

- What could go wrong?



# Reporting Regression Results

## Essential components:

- **Sample size:**  $n = 250$  observations
- **Model equation:**  $\hat{Y} = b_0 + b_1X$
- **Coefficient estimates:**
  - Point estimates
  - Standard errors
  - 95% confidence intervals
- **Statistical significance:** t-statistics, p-values
- **Model fit:**  $R^2$ , residual standard error
- **Interpretation in context:**
  - What does the slope mean practically?
  - Clinical/scientific significance vs statistical
- **Diagnostics:** Did assumptions hold?

**Example:** “For each 1-unit increase in BMI, systolic BP increased by 1.2 mmHg (95% CI: 0.8-1.6,  $p < 0.001$ ). The model explained 23% of the variance in SBP ( $R^2 = 0.23$ ).”

# Common Mistakes to Avoid

## Don't:

- Confuse correlation with causation
  - “X is **associated with** Y” not “X **causes** Y”
- Over-interpret R-squared
  - Low  $R^2$  doesn't mean the relationship isn't important
- Extrapolate beyond the data range
  - Don't predict for X values outside your sample
- Ignore influential points without investigation
- Report only p-values without effect sizes
  - **Always** report confidence intervals!
- Forget to check assumptions
  - Always create diagnostic plots
- Use jargon without explanation
  - Explain terms for non-statistical audience

# Creating Effective Visualizations

## Best practices:

- **Always label axes** with units
- **Include informative titles**
- **Use appropriate plot types**
  - Scatterplot for continuous X and Y
  - Boxplot for categorical X, continuous Y
- **Show the data and the model**
  - Points + regression line
  - Confidence/prediction bands
- **Use color wisely**
  - Colorblind-friendly palettes
  - Sufficient contrast
- **Keep it simple**
  - One main message per plot
  - Remove unnecessary elements

## Example: Publication-Quality Plot

```
# Create professional scatterplot with regression
model <- lm(sbp ~ age, data=health)

# Set up plot
plot(health$age, health$sbp,
     xlab="Age (years)",
     ylab="Systolic Blood Pressure (mmHg)",
     main="Relationship between Age and Blood Pressure",
     pch=16, col=rgb(0,0,0,0.3), cex=1.2)

# Add regression line
abline(model, col="red", lwd=2)

# Add confidence band
new_age <- seq(min(health$age), max(health$age), 0.1)
pred_ci <- predict(model,
                   newdata=data.frame(age=new_age),
                   interval="confidence")
lines(new_age, pred_ci[, "lwr"], col="blue", lty=2)
```

# Case Study: FEV and Age

**Research Question:** How does lung function (FEV) change with age in children?

**Data:** 654 children aged 3-19 years

- **Outcome:** FEV (forced expiratory volume, liters)
- **Predictor:** Age (years)

**Hypothesis:** FEV increases with age as lungs develop

## Analysis Plan:

- 1 Descriptive statistics and visualization
- 2 Fit simple linear regression
- 3 Check assumptions
- 4 Interpret results
- 5 Consider alternatives if needed

# Step 1: Descriptive Statistics

```
# Summary statistics
```

```
summary(fev_data$age)
```

```
#   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#   3.00   8.00   10.00   9.93  12.00   19.00
```

```
summary(fev_data$fev)
```

```
#   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#  0.791   1.981   2.547   2.637   3.119   5.793
```

```
sd(fev_data$age)      # 2.95 years
```

```
sd(fev_data$fev)      # 0.87 liters
```

```
# Correlation
```

```
cor(fev_data$age, fev_data$fev)
```

```
# r = 0.877 (strong positive correlation)
```

## Step 2: Initial Visualization

```
# Scatterplot
plot(fev_data$age, fev_data$fev,
     xlab="Age (years)",
     ylab="FEV (liters)",
     main="FEV vs Age in Children",
     pch=16, col=rgb(0,0,1,0.3))
```

# Observations:

- # - Strong positive relationship
- # - Possible non-linearity (steeper for younger)
- # - Greater variability for older children

**Initial impression:** Linear model may work, but check diagnostics

## Step 3: Fit Linear Model

```
# Fit simple linear regression
model <- lm(fev ~ age, data=fev_data)
summary(model)

# Coefficients:
#               Estimate Std. Error t value Pr(>|t|)
# (Intercept)  0.43161    0.06328   6.821  1.8e-11 ***
# age          0.22204    0.00608  36.515  < 2e-16 ***
# ---
# Residual standard error: 0.358 on 652 degrees of freedom
# Multiple R-squared:  0.6714,    Adjusted R-squared:  0.6709
# F-statistic: 1333 on 1 and 652 DF,  p-value: < 2.2e-16

# Interpretation:
# FEV increases by 0.222 liters per year of age
# Model explains 67.1% of variance
```



## Step 4: Check Assumptions

```
# Diagnostic plots
```

```
par(mfrow=c(2,2))
```

```
plot(model)
```

```
par(mfrow=c(1,1))
```

```
# Observations:
```

```
# 1. Residuals vs Fitted: slight curve (non-linearity?)
```

```
# 2. Q-Q plot: reasonably linear (normality OK)
```

```
# 3. Scale-Location: some funneling (heteroscedasticity?)
```

```
# 4. Residuals vs Leverage: a few high leverage points
```

```
# Potential issues:
```

```
# - Non-linearity (could try quadratic or log)
```

```
# - Non-constant variance (more spread for older ages)
```

## Step 5: Consider Alternatives

### Given diagnostic concerns, try quadratic model:

```
# Quadratic model
model_quad <- lm(fev ~ age + I(age^2), data=fev_data)
summary(model_quad)

# Coefficients:
#              Estimate Std. Error t value Pr(>|t|)
# (Intercept)  1.04423    0.10055   10.39 < 2e-16 ***
# age          -0.01626    0.02273   -0.72  0.475
# I(age^2)      0.01201    0.00109   11.00 < 2e-16 ***
# ---
# Multiple R-squared:  0.7198,    Adjusted R-squared:  0.7189

# Better R-squared (72% vs 67%)
# Check diagnostics again...
```

## Step 6: Interpret and Report

### Final Model: Quadratic relationship

$$\text{FEV} = 1.04 - 0.016 \times \text{Age} + 0.012 \times \text{Age}^2$$

### Key Findings:

- FEV increases with age in children, but non-linearly
- Rate of increase accelerates with age
- Model explains 72% of variance
- All coefficients highly significant ( $p < 0.001$ )

### Conclusion:

- Strong positive relationship between age and lung function
- Quadratic model fits better than linear
- Results consistent with biological lung development
- Caution: Don't extrapolate to adults (age  $> 19$ )

# Final Project Presentations

## Format:

- 10-12 minute presentation
- 3-5 minutes for questions
- PowerPoint, Beamer, or Google Slides

## Required content:

- 1 Introduction and research question
- 2 Data description and summary statistics
- 3 Statistical methods used
- 4 Results (tables, figures)
- 5 Interpretation and conclusions
- 6 Limitations

## Grading criteria:

- Clarity and organization (20%)
- Appropriate statistical methods (30%)
- Correct interpretation (25%)
- Quality of visualizations (15%)

# Tips for Successful Presentations

## Do:

- Practice your presentation multiple times
- Speak clearly and at moderate pace
- Make eye contact with audience
- Explain statistical terms
- Focus on interpretation, not just methods
- Use clear, uncluttered slides
- Tell a story with your data

## Don't:

- Read directly from slides
- Use too much text on slides
- Skip over assumptions or limitations
- Ignore questions you don't know – say “good question, I'll look into that”
- Go over time limit

## Avoid these mistakes:

- 1 **Too much technical detail**
  - Focus on interpretation, not formulas
- 2 **No context for numbers**
  - Always interpret in practical terms
- 3 **Cluttered visualizations**
  - One clear message per plot
- 4 **Ignoring limitations**
  - Acknowledge what your analysis can't tell you
- 5 **Poor time management**
  - Practice with a timer
- 6 **Not checking assumptions**
  - Always show diagnostic plots

## This week we covered:

- **Regression diagnostics:**
  - Four key assumptions (LINE)
  - Diagnostic plots
  - Identifying influential points
- **Transformations:**
  - Log, square root, polynomial
  - When and why to transform
- **Inference:**
  - Hypothesis tests for slope
  - Confidence and prediction intervals
- **Communication:**
  - Reporting results clearly
  - Creating effective visualizations
  - Using AI to enhance presentations

**Final Project Presentations:** Next week!

**Prepare:** Practice your presentation, check all analyses, create clear

Thank you!

**Good luck with your final projects!**

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